

**Integrating Virtual Reality with Computational Fluid Dynamics for Blood Flow
Assessment in Vascular Disorders: Year 2**

Abstract

Vascular disorders, such as cerebral aneurysms and carotid stenosis, impact nearly half of the US population. Of these, 1-3% of patients suffer stroke, significantly increasing patient mortality. Clinical treatment of vascular disorders entails stent implantation via endovascular surgery, a highly invasive therapy that can pose significant risk to patients.² Therefore, there is a need for selective treatment based on the severity of the disease (i.e., risk of stroke). Use of angiographic injection, the clinical standard for diagnosis of vascular disorders, is highly qualitative and often inconclusive of disease severity (i.e., risk of stroke). Application of computational fluid dynamics (CFD) can enable quantifying blood flow (hemodynamics) from an angiography, allowing for more informed prediction of disease severity. However, widespread use of CFD in the medical sphere is hampered by significant barriers such as high costs, technical complexity, and need for extensive computational power. Towards a broader adoption of CFD in diagnosis of vascular disorders, Year 1 of this study developed a low-cost, user-friendly, lightweight, open-source web application to visualize hemodynamic patterns in vascular disorders. Despite promising outcomes in visualization of post-CFD blood flow patterns, viewing hemodynamic data on a 2D computer screen masked critical information, and therefore, made analysis of hemodynamic patterns challenging. To solve this problem, Year 2 of this study focused on developing a virtual-reality (VR) application to view CFD-derived hemodynamic data in a highly interactive 3D environment. Rigorous coding using C# and associated libraries was performed in Unity to process hemodynamic input files and generate quantitative images and video plots on a VR headset. Analysis metrics including velocity vector magnitude, wall shear stress, time average, and oscillatory shear index were displayed to aid clinicians in accurately diagnosing the severity of the vascular disorder. The interactive application also allows data manipulation (e.g., panning, zooming) in 3D to highlight specific arterial regions to clinicians. Overall, the VR application developed in this work improves the accuracy of hemodynamic pattern recognition, and thereby, significantly enhances the diagnostic potential of CFD to assess the severity of vascular disorders.

Introduction

Vascular diseases are among the leading causes of death in the United States. These diseases include cerebral aneurysms, a bulge or ballooning in a blood vessel in the brain and carotid stenosis, the blockage or narrowing of the carotid artery (Mayo Foundation for Medical Education and Research, 2023). Diagnosis of these diseases is complex, and treatment often involves highly invasive surgery with relatively low chances of success (Geers et al., 2011). Due to the large degree of uncertainty surrounding the mortality, rupture risk, and severity of vascular diseases, accurate diagnosis is key in order to minimize risk and avoid unnecessary treatment.

Diagnosis of vascular disease relies on angiographic injection, a process which utilizes a contrast medium coupled with an angiogram to view the hemodynamics (blood flow) in an artery (Moore et al., 2017). Further study of such hemodynamics through Computational Fluid Dynamics (CFD) can quantitatively diagnose cerebral aneurysms and their need for treatment. However, CFD provides 3D data that is difficult to analyze and interpret for clinicians (Frösen, 2016). Studies have developed quantitative parameters for easier diagnosis, but such procedures require significant computational power and resources that most clinicians lack (White et al., 2023).

Virtual-Reality (VR) is a rising technology with the potential for 3D visualization of hemodynamic (blood flow) patterns for diagnosis of vascular disease and the development of an accurate treatment plan (Venn et al., 2023). The VR environment can display quantitative data such as Wall Shear Stress (WSS), Velocity Vector Magnitude (VVM), and Oscillatory Shear Index (OSI) through 3D plots. Therefore, a clinician-friendly VR application will address the need for better visualization technologies towards developing safer responses to vascular disease.

Literature Review

Diagnosis

The primary method for diagnosis of vascular diseases is the angiographic injection. The method entails the injection of a contrast medium into the bloodstream and the subsequent monitoring of how the dye travels through vasculature with an MRI-like scan (Abdollahi et al., 2021). Analysis of 2D/3D angiographic imaging yields a black-and-white animation of the contrast medium. The image/video can be used to identify aneurysms, stenosis, and other vascular conditions; however, concluding the rupture risk of the condition is difficult with strictly qualitative information of blood flow (White et al., 2023). Because of this, the bulk of diagnosis for vascular disease lies with the clinician, resulting in an error-prone process. Lack of treatment of vascular diseases damages the endothelial cells in the arterial wall, resulting in a weak structure and potential for rupture (Gimbrone, 1999). The need for accurate diagnosis will be exemplified through the cases of cerebral aneurysms and carotid stenosis.

A cerebral aneurysm (CA) describes the bulging or ballooning of an artery in the brain. CAs are known to impact those around the age of 50, with a larger connection to males over females. An estimated 6.7 million people in the US have an unruptured cerebral aneurysm, but most are undiagnosed due to the lack of symptoms and the complex diagnosis methods of the disease (Brain Aneurysm Foundation, 2023). Only 1-2% of cerebral aneurysms rupture each year, but ruptured aneurysms can cause hemorrhage and patient death if improperly treated. Treatment of cerebral aneurysms involves an intracranial angioplasty, subjecting patients to the risks of brain surgery and creating a need for accurate diagnosis (Geers et al., 2011).

Carotid stenosis characterizes the blockage or narrowing of the artery in the neck. The primary cause of carotid stenosis is atherosclerosis, the buildup of lipids in/around the artery

(Song et al., 2020). Over 50 million people worldwide are impacted by carotid stenosis. Treatment of carotid stenosis is primarily done through the placement of a stent, a highly invasive surgical procedure. However, there is documented evidence of the possibility of stent failure and patient death post-surgery. Therefore, similar to cerebral aneurysms, accurate diagnosis is imperative to developing a comprehensive treatment plan to carotid stenosis.

Computational Fluid Dynamics

Computational Fluid Dynamics is a physics-based approach to increasing the quantitative data available to clinicians for a more informed diagnosis approach (Lee, 2011). By employing the Navier-Stokes Equations and complex equation-solvers, quantitative hemodynamic (blood flow) data can be derived from a 3D rotational angiography (Jeong & Rhee, 2012; Yagi et al., 2020). The data can be in the form of Velocity Vector Magnitude (VVM), Wall Shear Stress (WSS), and Oscillatory Shear Index (OSI) (Han, 2021).

VVM describes the velocity of specific points in the artery. Narrowed regions of the artery would have higher VVM, placing more stress on the wall, while expanded regions would have lower VVM, symbolizing inadequate blood flow (White et al. 2023). WSS represents the shear stress placed on the wall, serving as a direct measure for rupture risk. A higher WSS in narrowed regions of the artery (stenosis) signifies a higher rupture risk. OSI explains the reversal of blood flow in the artery, caused by abnormal arterial shapes. OSI applies most for aneurysm diagnosis as the expanded arterial wall causes the reversal of blood flow, placing stress on the arterial wall and providing inadequate blood flow.

These hemodynamic parameters have several applications in the diagnosis of cerebral aneurysms. The first application of hemodynamic parameters is in routine diagnosis. Clinicians are able to use hemodynamic parameters to estimate the rupture risk of vascular diseases, as

stated before. WSS is most applicable to stenosis, while OSI is preferred in aneurysm cases. The second application of hemodynamic parameters is the prediction of vascular disease formation. Studies have used hemodynamic parameters for pre-diagnosis, anticipating the formation of cerebral aneurysms (Mantha et al., 2006). Such research demonstrates a preventive care application for hemodynamic analysis, especially in patients with genetic predisposition to the disease. The third application of hemodynamic parameters is pre/post treatment evaluations. Measuring hemodynamic parameters can provide a quantitative lens into the success of the treatment and the need for further post-treatment monitoring (Li et al., 2021).

Virtual Reality

The primary concern with using CFD is the difficulty in visualizing hemodynamic data. CFD data is inherently three-dimensional (3D) and therefore difficult to visualize on a two-dimensional (2D) screen. Viewing hemodynamics on a flat screen masks information and results in difficulty in understanding hemodynamic patterns to characterize disease formation. The causes of information reduction include a lack of comparison across the cardiac cycle, a crucial indicator of potential disease; a loss of supplementary data, such as biomedical responses (often unpredictable); and inefficient pattern recognition, caused by a large volume of data in a limited 2D display (Venn et al., 2023).

Virtual Reality (VR) can increase the diagnosis potential of CFD for vascular diseases by resolving the issues found in conventional CFD visualization methodology (Mourtzis et al., 2022). VR technology can increase the interactivity with the data using depth cues and intuitive interactivity with points (Venn et al., 2023).

Software and Purpose

Based on the information discussed, it is clear that the use of VR technology can address several limitations of CFD in diagnosis of vascular diseases. To broaden access to such technology, an open-source easy-to-use VR application can be developed (Venn et al., 2023). The following section will discuss the specific software requirements for the application.

Visualization of hemodynamics in an artery consists of two main parts: the vector points representing hemodynamic parameters (VVM, WSS, and OSI) and a low-opacity mesh derived from an STL file to represent the arterial wall. Combining these two parts results in a comprehensive view of the artery and its hemodynamics in localized areas. Execution of VR development is commonly done using a game engine such as Unity which provides a unified way to deploy an application on multiple platforms (Korzeniowski et al., 2018). Unity also allows for community-access to projects, facilitating a greater reach (Mourtzis et al., 2022).

Methods

To generate virtual-reality plots, the following parameters were employed: velocity vector magnitude (VVM), wall shear stress (WSS), time average wall shear stress (TAWSS), and oscillatory shear index (OSI). The following section discusses the calculation process for each of the parameters:

1. VVM was calculated by combining the x-velocity (u), y-velocity (v), and z-velocity (w)

in the equation $VVM = \sqrt{u^2 + v^2 + w^2}$. The u, v, and w values were calculated using Ansys Fluent.

2. WSS was derived using the VVM near the wall of the artery which involves complex calculations done by Ansys Fluent.

3. TAWSS was computed using the average WSS at a given point over the cardiac cycle, highlighting the absolute values of WSS in given areas.
4. OSI was determined through the following equations using the x, y, and z values for WSS over the cardiac cycle.

$$\text{OSI} = 0.5 \left(1 - \frac{A}{B} \right)$$

$$A = \int_{t_0}^{t_1} \left| \vec{\tau}_{\text{wall}} \cdot \hat{i} + \vec{\tau}_{\text{wall}} \cdot \hat{j} + \vec{\tau}_{\text{wall}} \cdot \hat{k} \right| dt$$

$$B = \int_{t_0}^{t_1} \left(\left| \vec{\tau}_{\text{wall}} \cdot \hat{i} \right| + \left| \vec{\tau}_{\text{wall}} \cdot \hat{j} \right| + \left| \vec{\tau}_{\text{wall}} \cdot \hat{k} \right| \right) dt$$

Generating 3D plots based on the parameters above involves a series of complex steps. Data from a .csv/.txt file is read and processed into a series of dictionaries. From there, data is sorted based on the parameter required for generation (e.g., VVM, WSS). Data from each point is used to instantiate (clone) a new sphere based on the x, y, and z values. The required parameter and color values are passed to the sphere for a looping animation. The animation interpolates between frames to increase smoothness and add additional data points.

To customize plot generation, scaling of plots was done using the transform functions of GameObjects. Furthermore, customized column input was developed through a series of global variables and sliders. Locomotion of the player was enabled through controllers and real-world movement.

References

- Abdollahi, R., Vahidi, B., Shojaei, P., & Karimi, M. (2021). A comparative study between CFD and FSI hemodynamic parameters in a patient specific giant saccular cerebral aneurysm. *AUT Journal of Modeling and Simulation*, 53(1), 23-38.
<https://doi.org/10.22060/miscj.2020.18742.5222>
- Brain Aneurysm Foundation (2023, August 28). *Statistics and Facts*.
<https://www.bafound.org/statistics-and-facts/#:~:text=Statistics%20and%20Facts%201%20Number%20of%20>
- Frösen, J. (2016). Flow dynamics of aneurysm growth and rupture: Challenges for the development of computational flow dynamics as a diagnostic tool to detect rupture-prone aneurysms. *Acta Neurochirurgica Supplement*, 89-95.
https://doi.org/10.1007/978-3-319-29887-0_13
- Geers, A. J., Larrabide, I., Radaelli, A. G., Bogunovic, H., Kim, M., Gratama van Andel, H. A., Majoie, C. B., VanBavel, E., & Frangi, A. F. (2011). Patient-specific computational hemodynamics of intracranial aneurysms from 3D rotational angiography and CT angiography: an in vivo reproducibility study. *AJNR. American journal of neuroradiology*, 32(3), 581–586. <https://doi.org/10.3174/ajnr.A2306>
- Gimbrone M. A., Jr (1999). Vascular endothelium, hemodynamic forces, and atherogenesis. The *American Journal of Pathology*, 155(1), 1–5.
[https://doi.org/10.1016/S0002-9440\(10\)65090-0](https://doi.org/10.1016/S0002-9440(10)65090-0)
- Han, P., Jin, D., Wei, W., Song, C., Leng, X., Liu, L., ... & Li, X. (2021). The prognostic effects of hemodynamic parameters on rupture of intracranial aneurysm: a systematic review and

- meta-analysis. *International Journal of Surgery*, 86, 15-23.
<https://doi.org/10.1016/j.ijisu.2020.12.012>
- Jeong, W., & Rhee, K. (2012). Hemodynamics of cerebral aneurysms: computational analyses of aneurysm progress and treatment. *Computational and mathematical methods in medicine*, 2012, 782801. <https://doi.org/10.1155/2012/782801>
- Korzeniowski, P., White, R. J., & Bello, F. (2017). VCSim3: A VR simulator for cardiovascular interventions. *International Journal of Computer Assisted Radiology and Surgery*, 13(1), 135-149. <https://doi.org/10.1007/s11548-017-1679-1>
- Lee B. K. (2011). Computational fluid dynamics in cardiovascular disease. *Korean circulation journal*, 41(8), 423–430. <https://doi.org/10.4070/kcj.2011.41.8.423>
- Li, G., Wang, H., Zhang, M., Tupin, S., Qiao, A., Liu, Y., ... & Anzai, H. (2021). Prediction of 3D Cardiovascular hemodynamics before and after coronary artery bypass surgery via deep learning. *Communications biology*, 4(1), 99.
<https://doi.org/10.1038/s42003-020-01638-1>
- Mayo Clinic. (2023, March 7). *Brain aneurysm - Symptoms and causes*.
<https://www.mayoclinic.org/diseases-conditions/brain-aneurysm/symptoms-causes/syc-20361483>
- Moore, D. (2017, July 9). *What is a cerebral angiography?* Healthline.
<https://www.healthline.com/health/cerebral-angiography>
- Mourtzis, D., Angelopoulos, J., & Panopoulos, N. (2022). Challenges and opportunities for integrating augmented reality and computational fluid dynamics modeling under the framework of industry 4.0. *Procedia CIRP*, 106, 215-220.
<https://doi.org/10.1016/j.procir.2022.02.181>

- Song, P., Fang, Z., Wang, H., Cai, Y., Rahimi, K., Zhu, Y., ... & Rudan, I. (2020). Global and regional prevalence, burden, and risk factors for carotid atherosclerosis: a systematic review, meta-analysis, and modelling study. *The Lancet Global Health*, 8(5), e721-e729. [https://doi.org/10.1016/S2214-109X\(20\)30117-0](https://doi.org/10.1016/S2214-109X(20)30117-0)
- Venn, J., Larkee, C. E., Garcia, G. J. M., Rayz, V. L., & LaDisa, J. F., Jr (2023). A workflow for viewing biomedical computational fluid dynamics results and corresponding data within virtual and augmented reality environments. *Frontiers in medical technology*, 5, 1096289. <https://doi.org/10.3389/fmedt.2023.1096289>
- White, R., Shields, A., Nagesh, S., Smith, E., Davies, J., Bednarek, D. R., Rudin, S., Ionita, C. N., & Chivukula, V. K. (2023). Investigating angiographic injection parameters for cerebral aneurysm hemodynamic characterization using patient-specific simulated angiograms. *Medical Imaging 2023: Biomedical Applications in Molecular, Structural, and Functional Imaging*, 39. <https://doi.org/10.1117/12.2653871>
- Yagi, S., Sasaki, T., Fukuhara, T., Fujii, K., Morita, M., Suyama, Y., ... & Hisatome, I. (2020). Hemodynamic analysis of a microanastomosis using computational fluid dynamics. *Yonago Acta Medica*, 63(4), 308-312. <https://doi.org/10.33160/yam.2020.11.013.t>

Research Plan

Name	Eshan Vipuil
School	West Shore Junior/Senior High School
Category	Mathematical and Computational Sciences
Research Teacher	Allison Vautier

A. Title

Integrating Virtual Reality with Computational Fluid Dynamics for Blood Flow Assessment in Vascular Disorders: Year 2

B. Question or Problem Being Addressed

There is a need for quantitative methods to visualize Computational Fluid Dynamics data in a more effective manner for diagnosis and increased understanding of cerebral aneurysms.

C. Hypothesis/Engineering Goals

1. Development of a VR (Virtual-Reality) application that can create 3D Vector Plot Models
2. Development of a MR (Mixed-Reality) application that can create 3D Vector Plot Models
3. Creation of a GUI that can be used to modify the 3D Vector Plot Models
4. Integration of the existing web platform with the VR/MR application

D. Criteria

1. The web app must be able to change color, scale, and other parameters of vector points
2. The web app must be able to accept data from various sources to create the vector plot
3. The program must provide a successful output at least 95% of the time

E. Rationale - Brief synopsis of the background that supports your research problem and explain why this research is important scientifically, and if applicable, explain any societal impact of your research.

Vascular Disorders are a major issue in the health of the people between the ages of 30 and 60, and affect nearly 20% of the US population. Although most cases lack severity and are asymptomatic, vascular rupture can quickly become life-threatening through hemorrhage and stroke. For this reason, there is a need to accurately diagnose the severity of vascular disorders. Currently the gold

standard for aneurysm diagnosis is an angiogram which provides a 3D image of the artery with an MRI-like scan. This scan only provides qualitative data which puts a great degree of analytical pressure on the clinician.

In order to combat this problem, techniques such as Computational Fluid Dynamics (CFD) are being developed to provide more quantitative data to a clinician by applying fluid flow principles to hemodynamics. Data from the angiographic injection can be better conceptualized through a 3D Vector or Wall Shear Stress Plot. Despite this, CFD technology has limitations in the present. Understanding a complex 3D simulation on a flat-screen is difficult, and can provide little value in real-world scenarios. Mixed Reality is a modern technology that allows for the visualization of 3D graphics alongside the real world. Use of such methods can provide a great deal of assistance in surgical procedures and diagnosis. In addition, patient information applications are also significant. Creation of a plot in 3D would increase quantitative information given to clinicians towards fewer cases of misdiagnosis.

- F. Materials List - List of all items used in research. Make sure to include concentrations of all chemicals, source and amount of all living organisms, and all equipment used. The list should be bulleted or numbered.

Item	Source
Computer	Researcher
Meta Quest 2 Headset	Florida Institute of Technology

G. Procedures

Safety Plan

1. Breaks will be taken when using the VR Headset for periods of time greater than 20 minutes per the American Academy of Ophthalmology
2. No audio will be utilized in the application

Experimentation Steps (Safety needs to be included throughout the experimentation steps)

1. Develop system to visualize vector magnitude, wall shear stress, and the arterial wall in a closed VR environment through the Unity Game Development Software (done at home)
2. Create mathematical parameters/formulas to estimate rupture probability from hemodynamic data - statistical analysis
3. Combine the two systems to create a virtual-reality application for viewing of vascular disorders in 3D.
4. Test the completed prototype with patient geometries (<http://ecm2.mathcs.emory.edu/aneuriskweb/about>) to ensure 95% accuracy in image generation (experimentation at home).
5. Experimentation will be done at home under the supervision of the designated

supervisor (see Form 3). Florida Institute of Technology will be used as a remote work site with mentorship from Dr. Venkat Keshav Chivukula (will not directly supervise, but will provide mentorship).

Disposal (What are the specific procedures? Even if you aren't responsible for them). Disposal of chemicals and other materials will need to also be listed.

1. N/A

H. Data Analysis: Describe the procedures you will use to analyze the data that answers research questions, hypotheses, or engineering goals. Steps should be numbered.

1. Data will be obtained from a de-identified public source:
<http://ecm2.mathcs.emory.edu/aneuriskweb/about>
2. Success rate of the design will be calculated over 50 trials involving various patient geometries (see data source above)
3. Runtime estimates will be calculated to improve the efficiency of the program over iterations
4. Iteration of the prototype will occur until the desired 95% success rate is achieved

I. Bibliography - Specific citations needed depending on your project--human subjects, vertebrate animal care, microbiology protocol, etc. A total of five full citations are required for approval. Must be formatted in MLA/APA. See rule book for specific resources.

1. Frösen, J. (2016). Flow Dynamics of Aneurysm Growth and Rupture: Challenges for the Development of Computational Flow Dynamics as a Diagnostic Tool to Detect Rupture-Prone Aneurysms. In: Tsukahara, T., Pasqualin, A., Esposito, G., Regli, L., Pinna, G. (eds) Trends in Cerebrovascular Surgery. Acta Neurochirurgica Supplement, vol 123. Springer, Cham. https://doi.org/10.1007/978-3-319-29887-0_13
2. Geers, A. J., Larrabide, I., Radaelli, A. G., Bogunovic, H., Kim, M., Gratama van Andel, H. A., Majoie, C. B., VanBavel, E., & Frangi, A. F. (2011). Patient-specific computational hemodynamics of intracranial aneurysms from 3D rotational angiography and CT angiography: an in vivo reproducibility study. AJNR. American journal of neuroradiology, 32(3), 581–586. <https://doi.org/10.3174/ajnr.A2306>
3. Regeneron ISEF: A Program of Society for Science. (2024). <https://sspcdn.blob.core.windows.net/files/Documents/SEP/ISEF/2025/Rules/Book.pdf>
4. Vasant, I. (2021, June 14). Data visualization in virtual reality. Medium. <https://towardsdatascience.com/data-visualization-in-virtual-reality-32408475b66>

5. Venn, J., Larkee, C. E., Garcia, G. J. M., Rayz, V. L., & LaDisa, J. F., Jr (2023). A workflow for viewing biomedical computational fluid dynamics results and corresponding data within virtual and augmented reality environments. *Frontiers in medical technology*, 5, 1096289. <https://doi.org/10.3389/fmedt.2023.1096289>
6. White, R., Shields, A., Nagesh, S. V., Smith, E., Davies, J., Bednarek, D. R., ... & Chivukula, V. (2023, April). Investigating angiographic injection parameters for cerebral aneurysm hemodynamic characterization using patient-specific simulated angiograms. In *Medical Imaging 2023: Biomedical Applications in Molecular, Structural, and Functional Imaging* (Vol. 12468, pp. 245-263). SPIE.

SDS Citations: Cite all chemicals used here using the specific chemical's SDS form (not a general link to all SDS's) . Must be formatted in MLA/APA.They should also be found on Form 3.

1. N/A

Problem/Purpose

Vascular disorders, such as cerebral aneurysms and carotid stenosis, impact nearly half of the US population. Of these, 1-3% of patients suffer stroke, significantly increasing patient mortality. Clinical treatment of vascular disorders entails stent implantation via endovascular surgery, a highly invasive therapy that can pose significant risk to patients. Therefore, there is a need for selective treatment based on the severity of the disease (i.e., risk of stroke). Use of angiographic injection, the clinical standard for diagnosis of vascular disorders, is highly qualitative and often inconclusive of disease severity (i.e., risk of stroke). While application of computational fluid dynamics (CFD) can enable quantifying blood flow (hemodynamics) from an angiography, its clinical utility is hindered by significant barriers such as high cost, technical complexity, and need for extensive computational power. Year 1 of this project focused on developing a user-friendly lightweight web application to visualize post-CFD hemodynamic patterns. Despite promising results, viewing hemodynamic data on a 2D computer screen masked critical information. The goal of this project is to develop a virtual-reality (VR) application to view CFD-derived hemodynamic data in a highly interactive 3D environment towards improving hemodynamic pattern recognition and enhancing the diagnostic potential of CFD for the assessment of vascular disorder severity.

Engineering Goals

1. Develop a web app with the ability to generate 3D Vector Plot Models using Python 3.
2. Expand generation capabilities to VR using Unity and C#.
3. Visualize time-varying data through continuous animations in a VR environment.
4. Create a user interface to navigate through the VR scene using real-time movement and controller input.
5. Calculate and visualize key hemodynamic parameters - velocity vector magnitude, wall shear stress, and oscillatory shear index.

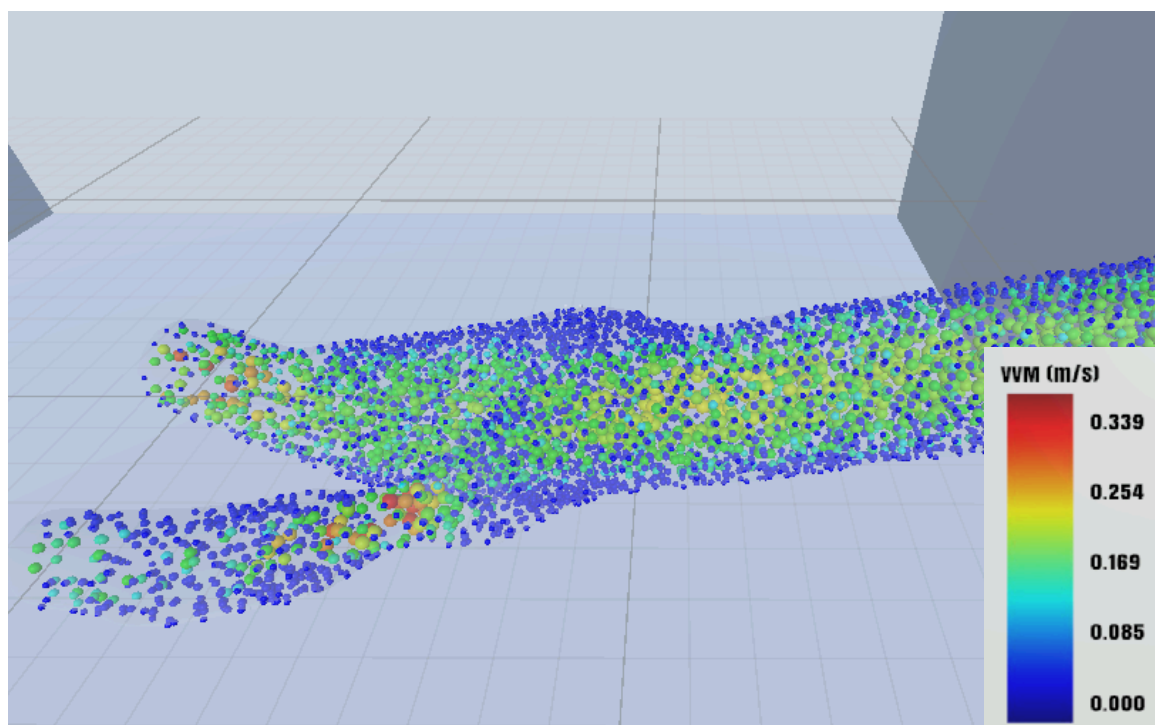
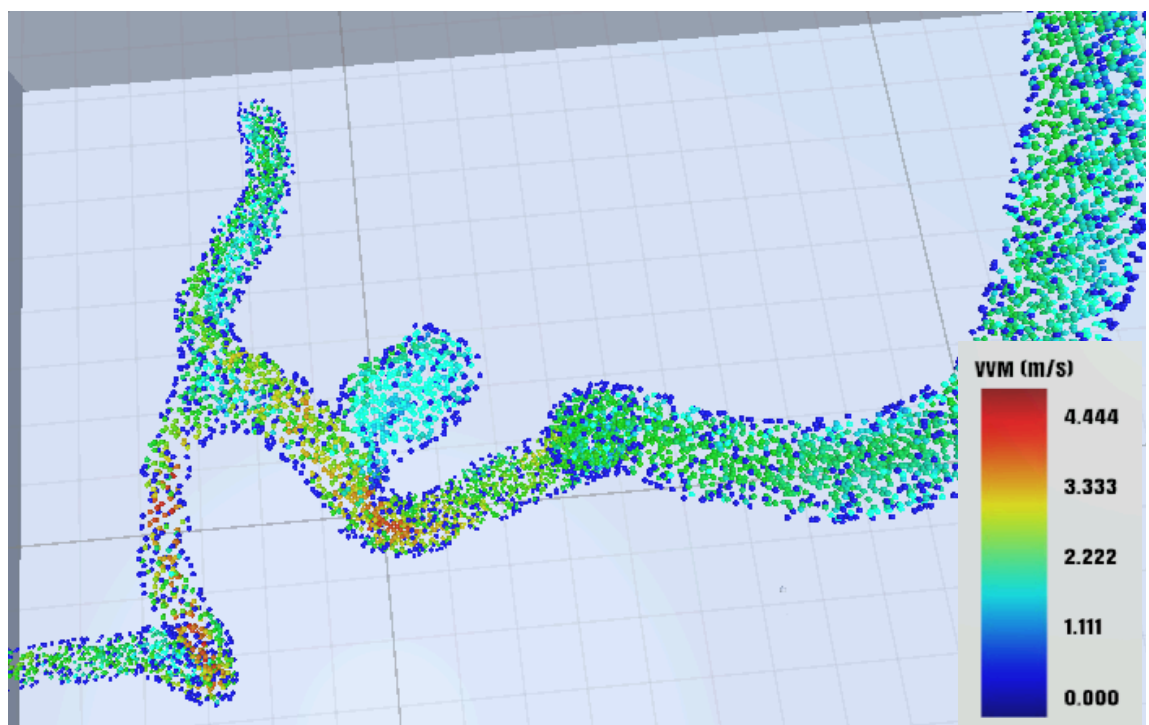
Criteria

1. The VR app must be able to change color, scale, and other parameters of points.
2. The VR app must be able to accept data from a spreadsheet to create the vector plot.
3. The program must provide a successful output at least 95% of the time.

Results

Figure 1

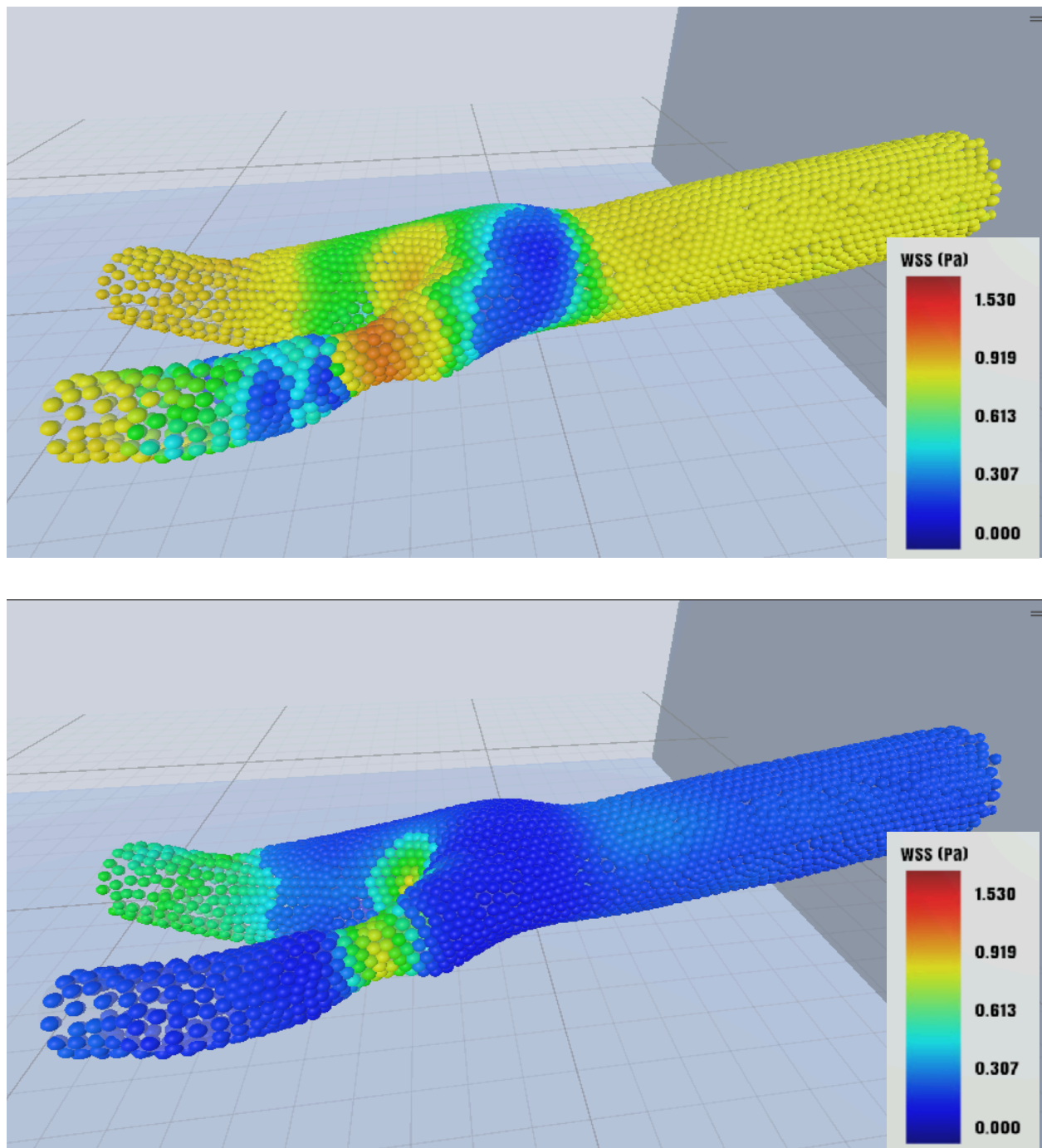
Image Generation of VVM In Aneurysm (top) and Carotid Stenosis (bottom)



Note. The images show the effect of increasing (aneurysm) and decreasing (stenosis) arterial cross-sectional area on velocity, explained by continuity principle ($A_1v_1 = c$).

Figure 2

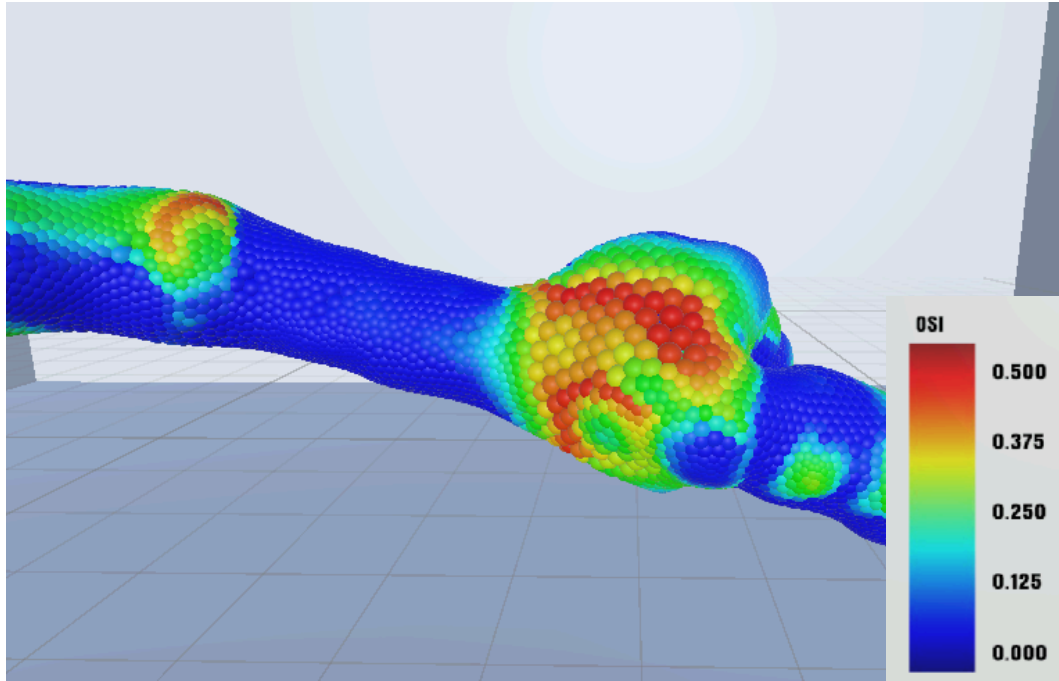
Wall Shear Stress in Carotid Stenosis: Systole (top) and Diastole (bottom)



Note. The stenosed region (highlighted by a red box) has a consistently higher WSS value than the surrounding regions in the systole (high blood flow) and diastole (low blood flow) stages, quantitatively highlighting the potential of stroke.

Figure 3

Oscillatory Shear Index in Aneurysm



Note. OSI displays the change in WSS. High OSI in the aneurysm region (highlighted by a red box) represents high rupture probability.

The final deliverable of the study was the creation of an interactive VR application. The application allows for user input of .csv and .txt files and the subsequent generation of a 3D environment using Unity and C# programming. The environment successfully displayed the intended hemodynamic parameters (i.e., velocity vector magnitude, wall shear stress, and oscillatory shear index) through a continuous 3D animation and an array of colors.

Velocity profile results are in line with the continuity principle ($A_1V_1 = A_2V_2$), wherein an increase in arterial cross-sectional area in case of an aneurysm yields a decrease in flow velocity

and vice versa for carotid stenosis. Wall shear stress changed proportionally with change in velocity, showing lower values in an aneurysm and higher values in carotid stenosis. Graphical analysis using box plots reveal changes in wall shear stress data during the cardiac cycle. Lastly, oscillatory shear index displays significant changes in wall shear stress in the aneurysm region, caused by turbulent flow. Although wall shear stress in an aneurysm region is low, the bulging of the artery causes high oscillatory shear index, representing rupture probability.

Conclusions

All five of the stated engineering goals were met. Velocity vector magnitude and wall shear stress were successfully visualized using 3D vector and scalar plots. Time varying data was compiled and animated to enable analysis of hemodynamic patterns over the cardiac cycle. A complete user interface allowed in-depth analysis of generated plots through features such as panning and zooming. Each of the visualizations mentioned can be traversed using the real-world movement of the user.

Each of the stipulated criteria were also met in the final application. The application allowed for changing of color, scale, and other parameters to deliver an interactive final image for quantitative hemodynamic analysis. Data was accepted from .csv or .txt files and the C# program resulted in an accurate output in more than 95% of the test cases.

Overall, the VR application developed in this project is a user-friendly platform that has the potential to significantly transform medical imaging and clinical diagnosis for vascular disorders.

Future Studies

In future work, the VR visualization platform can be further developed by integrating machine learning algorithms to enable automated analysis of hemodynamic data. This advancement coupled with other diagnostic parameters will improve the quantitative analysis and allow for more accurate prediction of disease progression and arterial rupture probability. Together, these developments will allow clinicians to make more informed decisions on an effective treatment plan to reduce mortality associated with vascular disorders.

In addition to integrating machine learning algorithms, the VR application can be enhanced through calculation of additional hemodynamic parameters such as time to peak and mean transit time to more accurately estimate arterial rupture probability.